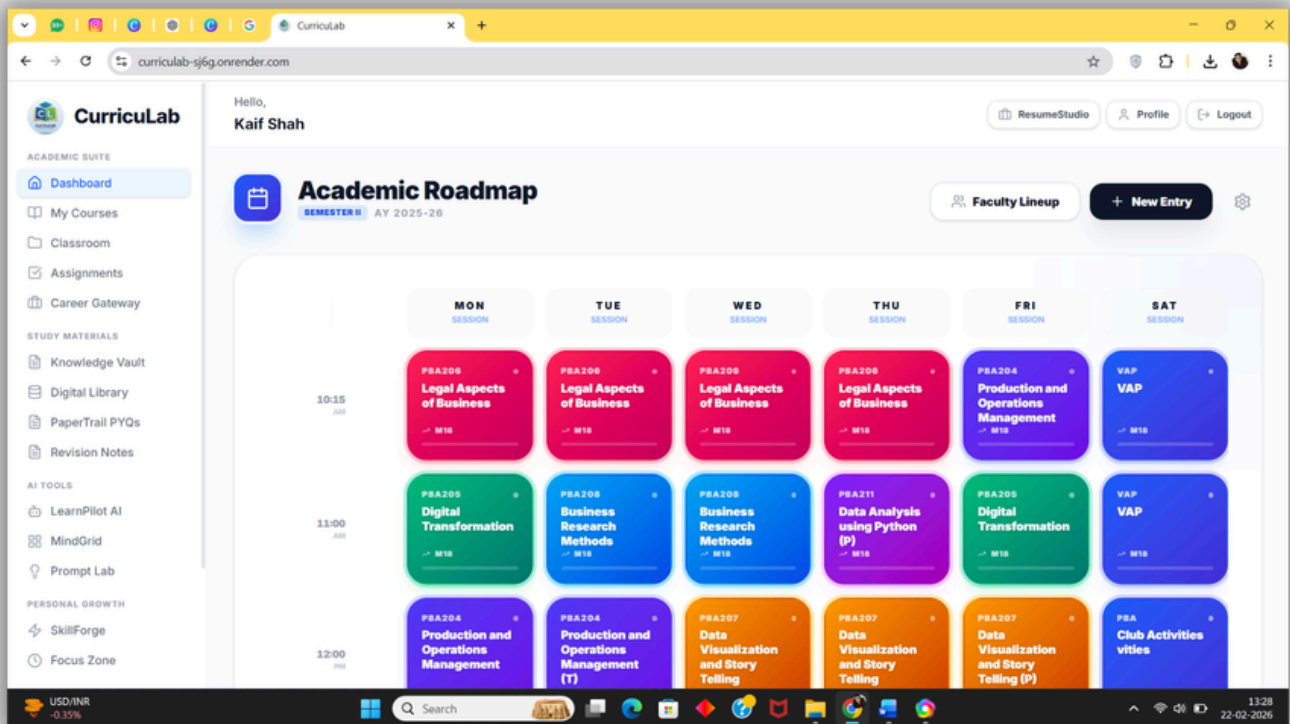


CurricuLab

The Student Centric Study Management System



The Research & Quality Team

Kaif Shah

Quality Assessment Manager

Akash Jayaprakash Mangalamthodi

R&D Manager

Introduction

The Research & Quality Team plays a central and strategic role in the successful development of CurricuLab – The Student Centric Study Management System. While the Data Acquisition Team is responsible for generating and collecting academic content, the Research & Quality Team ensures that all study materials meet academic standards, maintain consistency, and align strictly with syllabus requirements.

This team ensures that the platform remains reliable, structured, and examination-oriented. The primary goal is to maintain academic integrity, eliminate inconsistencies, and enhance the credibility of the entire system before the content is deployed to the technical backend

Vision and Purpose of Research & Quality Team

The vision of the Research & Quality Team is to build a high-quality academic content validation system that ensures:

- Accuracy of subject concepts
- Alignment with university syllabus
- Standardized formatting
- Clear and structured answers
- Elimination of AI hallucination
- Proper word-count adherence according to marks

The purpose is not just to review content but to create a structured academic quality control mechanism that strengthens the learning ecosystem

Roles and Responsibilities

Kaif Shah – Quality Assessment Manager

As the Quality Assessment Manager, the responsibilities include:

- Reviewing all study materials received from the Data Acquisition Team
- Verifying alignment with syllabus and book references
- Checking word count according to mark segmentation (2, 5, 10, 15 marks)
- Ensuring logical flow and clarity of answers
- Identifying contextual errors or AI hallucinations
- Maintaining formatting consistency across all subjects
- Coordinating corrections before final approval
- Final validation before handing content to the Technical Team

The Quality Manager ensures that no content is uploaded without strict academic validation.

Akash Jayaprakash Mangalamthodi – R&D Manager

The Research & Development Manager focuses on improving content generation systems and enhancing quality mechanisms.

Responsibilities include:

- Evaluating AI-generated content models
- Enhancing custom ChatGPT tools (NoteSmith, Syllabrix, CaseOrbit)
- Researching better academic structuring methods
- Ensuring real-world relevance in case studies
- Designing improved prompt frameworks
- Reducing AI-generated inconsistencies
- Supporting innovation in study material development

The R&D Manager ensures that content generation evolves continuously and improves in quality.

Research & Quality Workflow Process

Step 1 : Content Reception

- HTML files received from Data Acquisition Team
- Study notes, question banks, and case studies submitted

Step 2: Academic Review

- Cross-check with syllabus
- Verify concept clarity
- Ensure completeness of topics

Step 3 : Structural & Formatting Review

- Uniform headings and subheadings
- Proper paragraph segmentation
- Correct numerical formatting
- Consistent font and structure

Step 4: Quality Evaluation Parameters

- Accuracy of concepts
- Word count precision
- Logical explanation
- Removal of repetitive content
- Elimination of hallucination

Step 5 : Feedback & Correction

- Corrections communicated
- Revised version reviewed again

Step 6 : Final Approval

- Approved content handed to Technical Team for system integration



Handling AI-Generated Limitations

During study material creation using AI tools like ChatGPT and Gemini, several issues were identified:

- Contextual inconsistency
- Repetitive explanations
- Incorrect word counts
- Formatting problems
- Hallucinated case studies

To overcome these challenges, the team supported the implementation of customized AI systems:

Customized AI Tools Monitored:

- NoteSmith – Unit-wise structured study notes
- Syllabrix – Mark-based question banks
- CaseOrbit – Industry-based case studies

The R&D Manager continuously improves these systems to maintain academic quality.

Performance and Quality Metrics

To maintain measurable standards, the team tracks:

- 100% syllabus coverage
- Mark-based word accuracy
- Zero hallucinated responses
- Uniform formatting across subjects
- Error-free numerical representation
- Structured case study validation

These metrics help in maintaining continuous improvement

Collaboration with Other Teams

The Research & Quality Team works closely with:

Coordinator

- Reports quality status
- Aligns academic standards with project vision

Data Acquisition Team

- Reviews generated content
- Suggests corrections
- Maintains continuous communication

Technical Team

- Ensures correct HTML formatting
- Verifies structured deployment
- Maintains database consistency

This collaboration ensures seamless integration between research, content, and technical development.

Conclusion

The Research & Quality Team is a foundational pillar of CurricuLab. Through structured validation, continuous research, and academic monitoring, the team ensures that every piece of content uploaded into the system is accurate, relevant, structured, and student-focused.

Under the leadership of:

Kaif Shah – Quality Assessment Manager

Akash Jayaprakash Mangalamthodi – R&D Manager

The department guarantees that CurricuLab maintains high academic standards and long-term sustainability.